Carlos Romero Madrid

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Computational Physicist • HPC Software Engineer • Creative Technologist

Profile & Mission: Computational Physicist (Double Degree Physics & Mathematics, 1st of Class MSc Plasma Physics) and HPC Software Engineer at BSC with 5+ years experience. Bridging extreme-scale HPC clusters (MN5, Leonardo, LUMI) with physical computing, generative sound, and narrative data visualization for the **democratization of science**. Pairing with LLMs for high-velocity cross-disciplinary prototyping.

Featured Creative Computing & Data Projects

MNusic — Supercomputer Sonification
HPC TELEMETRY & AUDIO • WEB APP ■
Translates tier-0 cluster metrics (MN5, LUMI) into real-time harmonic soundscapes (Web Audio API) and dynamic particle visuals, humanizing petascale computation.

Pipa-Music-Ball — Bio-Kinetic Sound
PHYSICAL IOT & SENSORS • GITHUB ■
Embedded microcontroller sphere (ESP32, 6-DoF IMU) translating feline kinetic play into generative microtonal acoustic phrases—tangible and playful data embodiment.

Expo-Curator — Visual Corpus Analytics
CURATION STUDIO & STAGING • GITHUB ■
Open-source studio application for automated image sub-classification, booklet design, and exhibition wall staging for cultural and science dissemination spaces.

After Effects Lab — Procedural Motion
MOTION DESIGN & VIDEO • CINEMA REEL ■
Still photographic composition combined with procedural Adobe After Effects motion graphics to turn static scientific research into broadcast-ready visual essays.

Technical Matrix & Creative Capabilities

Creative Computing & Audio
- Tangible Data Interfaces (ESP32/IMU)
- Web Audio API Real-Time Synthesis
- HTML5 Canvas & Generative Systems
- Inspired by Domestic Data Streamers

Visual Media & Motion Design
- **Adobe After Effects** (Motion Graphics)
- **Still Photography** & Spatial Framing
- **Expo-Curator** Exhibition Studio
- Kinetic Typography & Color Grading

HPC, AI & Compute Systems
- **Tier-0 Supercomputers** (MN5, LUMI, Slurm)
- **Computer Vision & MediaPipe**
- Parallel C++, Fortran, OpenMP/MPI/CUDA
- **Agentic LLM Pipelines & Web ML**

Professional Trajectory & Academic Foundation

Barcelona Supercomputing Center (BSC-CNS) — HPC Scientific Software Engineer May 2023 — Present

- *Performance Team (May 2025–Present):* DestinationE & ClimateDT digital twins on MareNostrum 5; developing ECLand.
- *Fusion Team (May 2023–Apr 2025):* Optimized nuclear fusion simulation codes with OpenMP/MPI/CUDA; ALYA-MAGNET modeling.

National Renewable Energy Centre (CENER) — Data Scientist May 2022 — Sep 2022

- Built Python real-time telemetry data pipelines for characterisation and predictive maintenance of offshore wind plants.

Univ. of Seville (Research Scientist) & Fraunhofer FHR (Cognitive Radar) 2018 — 2022

- Particle-in-cell & MHD plasma simulations on HPC clusters; lectured Quantum Physics; non-planar radar antenna optimization.

Academic Rigor: Double BSc Physics & Mathematics • MSc Plasma (1st) • MSc Adv. Maths Univ. Seville • Marseille • UNED

- **Double BSc in Physics & Mathematics** (Univ. Seville, 2013-18, **3rd of Class**); **MSc Plasma & Fusion Physics** (Aix-Marseille, 2019-20, **1st of Class**, 15.43); MSc Advanced Mathematics (UNED, 2023-Present).

■ **Live Interactive Portfolio & Telemetry Synthesizer:** Experience real-time MareNostrum 5 soundscapes, ambient crossfader, and video cinema mode at: **https://carrommad.github.io/cool-cv**